

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 4<sup>th</sup> Semester of B. Pharm. Examination

University Theory Examination October-November 2017

### PH210 Organic Chemistry-II

Date: 04/10/2017, Wednesday,

Time: 01:30 p.m to 04:30 p.m

Maximum Marks: 80

#### Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

#### SECTION- I

##### Q 1 Attempt any TWO of the following:

- A What is Huckel's rule for aromaticity? Explain electrophilic aromatic substitution reaction in benzene with respect to nitration and halogenation. 10
- B Write answers of followings.(2X 5 marks)
- a) Differentiate Conventional Synthesis Vs. Microwave assisted Synthesis. 05
- b) Neighbouring Group Effect. 05
- C Write answers of followings.(2X 5 marks)
- a) Cycloaddition reaction. 05
- b) Principle, advantages and disadvantages of Microwave assisted synthesis with respect to green chemistry approach. 05

##### Q 2 Attempt any TWO of the following:

- A What is Green Chemistry? Explain the twelve principle of Green Chemistry. 10
- B Write a short note on Friedel Craft alkylation with mechanism and suitable examples. 10
- C Write a short note on followings. (2X 5 marks)
- a) Nanochemistry. 05
- b) Nucleophilic acyl substitution reaction. 05

#### SECTION- II

##### Q 3 Attempt any TWO of the following:

- A Write a short note on Kolbe Synthesis and Cannizzaro reaction with suitable mechanism and examples. 10
- B Give any three preparations and reactions of Phenol. 10
- C Write answers of followings.(2X 5 marks)
- a) Reaction and mechanism of Malonic Ester Synthesis. 05
- b) Chemical reactions of Naphthalene (any two). 05

##### Q 4 Attempt any TWO of the following:

- A Explain Hoffman Rearrangement for Amine with suitable mechanism and examples. 10
- B Give any three preparations and reactions of Ester. 10
- C Write answers of followings.(2X 5 marks) 10
- a) Reaction and mechanism of Diel's alder reaction.
- b) Give any three preparation of Ketone.